Replicating Walker et al. (2023) with IPEDS Data

Anticipatory impacts of the repeal of Roe v. Wade on female college applicants

- Walker, Wisniewski, Torres, and Sharma (2023), *Economics Letters*
  - Examine *anticipatory* impacts of restrictive abortion policy on applications to top U.S. colleges.
  - Use the Common Data Set and U.S. News & World Report rankings.
  - Assume the overturning of federal protections in *Roe v. Wade* was anticipated.
  - Find a decrease in the share of women applicants.
- Replicate with IPEDS data using a slightly larger sample.
  - Find similar, but smaller, effects.
  - Confirm that results are driven by top-50 schools and schools with a large share of out-of-state applicants.

- Integrated Postsecondary Education Data System (IPEDS)
    - surveys Title IV schools
    - *Admissions and Test Score* survey contains applicants by gender
    - *Directory information* gives school state to match with policy
    - *Residence and migration of first-time freshman* is proxy for in-state apps
- Policy data is from Table 1 in Walker et al. (2023)
- *U.S. News & World Report Rankings* archive from Andrew Reiter GitHub repo

Table 1: Summary Statistics Comparison: Mean (SD)

	Ban States	Control States
IPEDS DATA		
Pre-2022 Share of Freshman Female Applicants	0.55 (0.07)	0.52 (0.09)
Number of States in Sample	10	21
Number of Schools in the Sample	25	78
WALKER DATA		
Pre-2022 Share of Freshman Female Applicants	0.56 (0.07)	0.52 (0.08)
Number of States in Sample	8	19
Number of Schools in the Sample	19	52

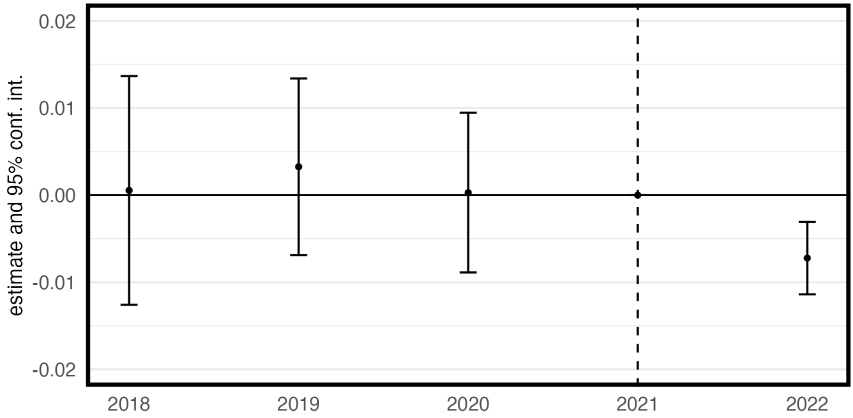
Two Way Fixed Effects Event Study

$$Y_{ijt} = \sum_{\substack{t=2018 \\ t\neq 2021}}^{2022} \left[\alpha_{ijt} \cdot Repeal_j \cdot Year(t)\right] + \tau_t + \delta_i + \varphi_j + \varepsilon_{ijt}$$

- Y_{ijt}: women share of applicants
  - school *i*
  - state *j*
  - year *t*
- $Repeal_j$: dummy indicating state *j* affected by repeal of Roe v. Wade
- τ_t, δ_i, φ_j: year, school, and state fixed effects
- ε_{ijt}: error term clustered on state

- α_{ijt}: effect of ban state relative to control relative to 2021

Figure 1: Effect on Share of Women Applications



Notes: 0.7^{***} percentage point decline (Walker et al find 0.9^{**})

Figure 2: Effect on Share of Women Applications for Schools ranked 51+

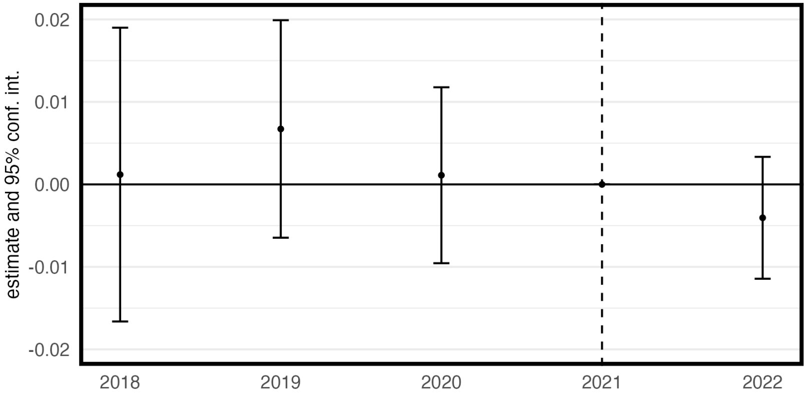
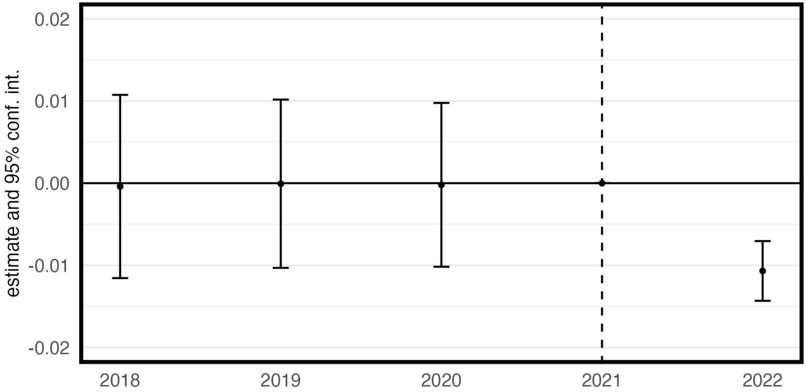
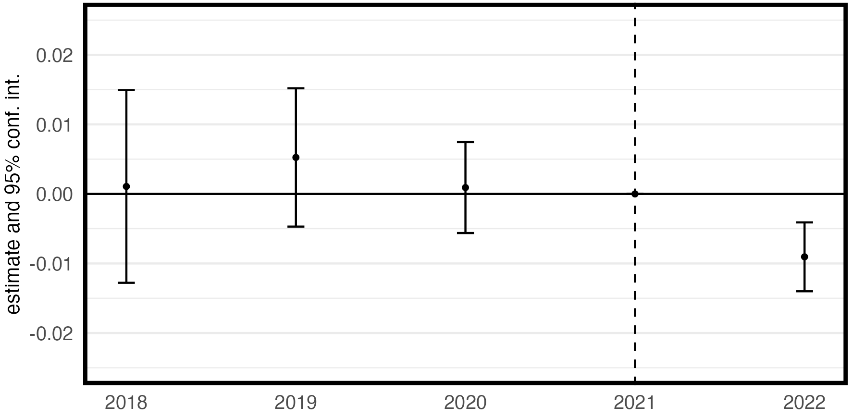


Figure 3: Effect on Share of Women Applications for top 50 Schools



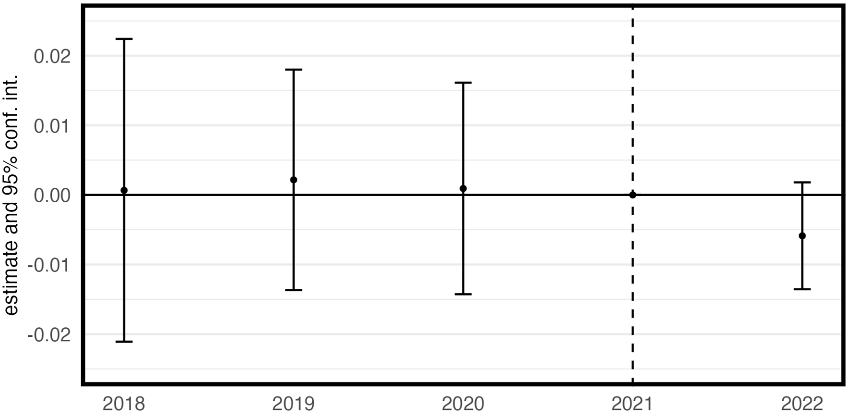
Notes: 1.06^{***} percentage point decline (Walker et al appears identical)

Figure 4: Effect on Share of Women Applications in Schools with $50\%+$ Out of State Freshmen



Notes: 0.9^{***} percentage point decline (Walker looks identical)

Figure 5: Effect on Share of Women Applications in Schools with $< 50\%$ Out of State Freshmen



<!–

Notes: 0.7^{***} percentage point decline (Walker et al find 0.9^{**})

- replicate Walker et al. (2023) with IPEDS data set –> slightly larger sample
- find more modest main effects but still significant decline
- limitations
    - only anticipatory effects
    - could be picking up substitution and "control" states might be affected
    - effects could be transitory
    - no time-varying covariates
- future directions
    - include time-varying covariates
    - extend data through present
        - staggered policy adoption
        - use modern estimation methods
    - use Meyer data with driving distance to nearest clinic instead of dummy

Folders

```
data/
    original data/
        IPEDS/                # contains ADM, EF-C, HD
        Table 1.xlsx         # extracted from Walker
        USNWR.xlsx           # GitHub rep
    saved data/
        maindf.Rdata
R
results/
    figures                   # saved with ggsave()
scripts/
    analysis/
        main.R                #all analyses in one script
    data processing/
        compileIPEDS.R
        outcome-policy-variables.R
```

Compile Data

```
#===== COMPILE ADM DATA ====
#get list of files in folder
folder.path <- here("data","original data","IPEDS","ADM")
lf <- list.files(here(folder.path))

#initalize dataframe
ll <- 1                                    #first file in list
df <- read_csv(here(folder.path,lf[[ll]]))   #read file
yr <- lf[[ll]] |> str_extract("\\d+")      #extract year from file name

df <- df |> select(UNITID,APPLCN,APPLCNM,APPLCNW) |> mutate(year = yr[[1]],.after = UNITID)
rm(yr,ll)

#loop over remaining files
for (ll in 2:length(lf)) {
  df0 <- read_csv(here(folder.path,lf[[ll]]))
  yr <- lf[[ll]] |> str_extract("\\d+")
  df0 <- df0 |> select(UNITID,APPLCN,APPLCNM,APPLCNW) |> mutate(year = yr,.after = UNITID)
  rm(yr)
  df <- df |> bind_rows(df0)
}

df <- df |> mutate(year = as.numeric(year))
rm(df0,ll)
```

Main Dataframe

```
#NOTE: efc was unit x year x state but
#summarise() to unit x year (with out.percent)
hd <- hd |>
  left_join(efc,join_by(UNITID,year),relationship = "many-to-many")

df <- adm |> left_join(hd,join_by(UNITID, year))

rm(adm,hd)
```

Example Analysis

```
#====== FUNCTIONS =====
tidymod <- function(mod) {
  mod |> tidy(conf.int = TRUE) |>
  transmute(term = as.numeric(gsub("\\D","",term)),
            estimate,conf.low,conf.high) |>
  add_row(term =2021,estimate = 0,conf.low = 0,conf.high = 0) }




#==== MAIN RESULT ====
#run model---
mod <- feols(wshare ~ i(year,ban,ref = 2021)|UNITID+STABBR+year,
             cluster = ~STABBR,DF)

#make figure---
##myplot() passes a tidy model into ggplot() with preferred themes
mod |> tidymod() |>  myplot() +
  scale_y_continuous(limits = c(-.02,.02),breaks = seq(-.02,.02,by = .01))
```